MATH1061 2022 T2+
Question 1.

Let $\mathcal{G}$ be the set of all graphs, and let $\preceq$ be a relation on $\mathcal{G}$ given by

$$G \preceq H \text{ if and only if } G \text{ is a subgraph of } H$$

for all graphs $G, H \in \mathcal{G}$.

(a) Prove that $\preceq$ is a partial order relation.
When typing your answer, you may use the character < instead of the symbol $\preceq$ wherever needed.

Transitivity
Consider the following for some arbitrary graphs A, B, C
if A ~ B and B ~ C, then A ~ C, such that A is a subgraph of C.
If A is a subgraph of B, then some collection of B's vertices and edges, say S, are contained within A's vertices, say T.
Hence, we could express this as:
S ⊆ T, and further more, T ⊆ K, where K is the set of vertices and edges in C.
Therefore, S ⊆ K, and hence, A ~ C.

Anti-symmetry
If A ~ B, and B ~ A, then A = B.
Consider again the edges and their respective vertices as sets - let A be the set S and B be the set T.
If A ~ B, and therefore, A is a subgraph of B, then S must be a subset of T, where T is the set for B.
If B ~ A, and therefore, B is a subgraph of A, then T must be a subset of S.
Hence, we have that S = T, and therefore, A = B.

Reflexivity
A ~ A exists for all A
Every graph is a subgraph of itself, so this is trivially proven.

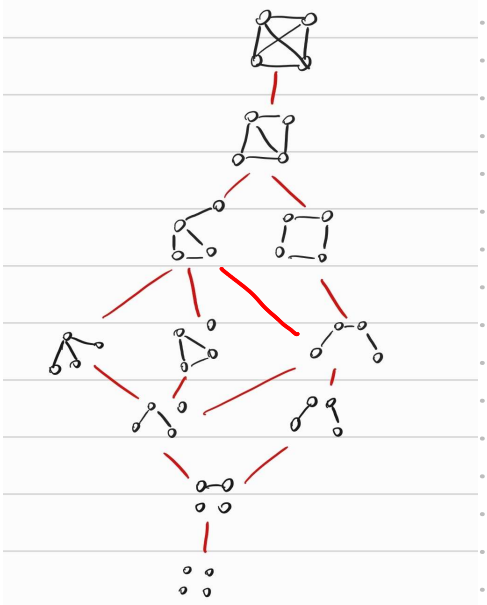
Let S_n be the set of all (unlabelled) graphs with n vertices.

(b) Briefly describe the greatest and least elements of the partially-ordered set $(S_5, \preceq)$, if they exist.

The least element in this case is the null graph - or a graph with no vertices and no edges.
Such a subgraph exists within every single graph, and no other subgraph can be composed of the null graph.

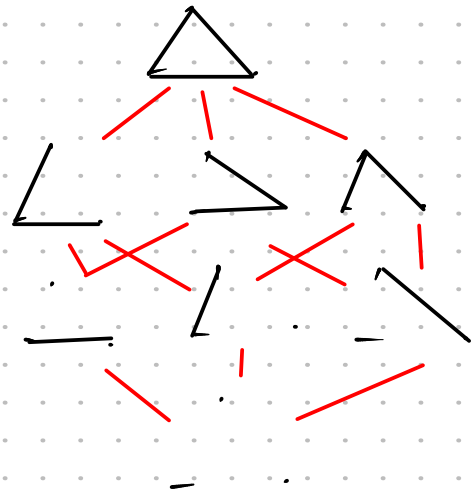
The greatest element would be a complete graph with all edges taken.
A graph that had more edges and more vertices than S(5) would no longer be a subgraph, and hence a limit is formed.

(c) Draw the Hasse diagram for the partially-ordered set $(S_4, \preceq)$. How many connecting lines does this Hasse diagram contain?
(Hint: $|S_4| = 11$.)



14 lines.

d) Consider that each "level" has ${}^{n}C_{k}$ items, where n is edges and k is levels. Consider the following example:-



At each stage, for every single edge present within the current graph, there are n amount of edges that you can remove. At each level, there is a combination of nCk items, as each level portrays how many edges have been removed.

In the above example, level one shows when 1 edge has been removed - and hence, it has 3C1 items. Level 2 shows when 2 edges have been removed, hence, there are 3C2 items.

For the edges between these items themselves, you must consider the amount of edges present within the original graph, as an indicator of how many edges can be removed. Hence, for a level, the formula to find the amount of lines is:
Edges that can be removed * Items that represent each removal
= e * nCk.
The e variable will decrement each stage you go down, where k will increment each stage you go down.
n represents the edges in the complete graph of however many variables you originally had.

Question 3.

Lara has invented a function f that takes multigraphs as input values and returns integers as output values, defined as follows:

For any multigraph G, we have $f(G) = |V(G)| \times |E(G)|$.

(a) Complete the following sentences:

The function $f : \{\text{all graphs}\} \to \mathbb{N}$ is neither injective nor surj... .

The function $f : \{\text{all connected graphs}\} \to \mathbb{N}$ is neither injective nor surj... .

Briefly justify your answers below.

For both graphs, it's simple to prove that both graphs are not injective and surjective.
For example, a graph that has 6 vertices and 1 edge, has the same output value as a 3 vertex and 2 edge graph.
For a connected graph, a 2 vertex 1 edge graph has the same as 1 vertex 2 edge graph.
Hence, by counter-example, they are not injective.
For a simple graph, and it's surjectivity, it cannot have multiple lines going between each graph.
Since a simple graph cannot have a loop, f(G) is impossible. Hence, not surjective.
Hence, the maximum amount of edges you can have is n(n-1)/2, which is a complete graph.
n(n-1)/2 = x has distinct restrictions, as first, plugging in an x value which does not compute for the quadratic equation shows a counter example
For a connected graph and it's surjectivity, consider that there is a distinct restraint such that there must be atleast n - 1
Hence, we have, at a minimum:
n(n - 1) = x, for some output value x.
We could sub in a value of x such that the quadratic equation does not have a solution.

(b) Lara thinks that $f(K_{m,n})$ is even for all $m, n \in \mathbb{Z}^+$. Are they correct? Carefully prove your answer.

The amount of vertices within a K(m, n) graph is m + n, which is not intrinisically even.
Consider |E(K(m, n))|.
nm(m + n)
Consider the case where both are even. It is clear nm(m + n) is also even.
Consider the case where only one is even:
2*k(2*j + 1)*(2*k + 2*j + 1)
All elements will turn out to be even.
Consider the case where both are odd:
(2*j + 1) * (2k + 1) * 2 * (j + k + 1)
Therefore, also even.
Hence for all three cases, the product is even.

Question 4. a)

Jordan is building a tower out of different types of blocks. The blocks come in two different sizes and many different colours. There is an unlimited supply of:

- blocks of height 1 that come in 2 different colours, and
- blocks of height 2 that come in 6 different colours.

Jordan considers two towers different even if they are reflections of one another. For example, the tower formed by placing a red block of height 1 on top of a red block of height 2 is different from the tower formed by placing a red block of height 2 on top of a red block of height 1.

(a) Let a_n be the number of different towers of height n that Jordan can construct. Then we have

$a_1 = 2,$

$a_2 =$ 6 + 2^2 ,

and

(a) Let a_n be the number of different towers of height n that Jordan can construct. Then we have

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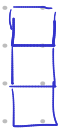
and

$$a_n = Xa_{n-1} + Ya_{n-2}$$

for all integers $n \geq 3$, where

$X =$ 2 and

$Y =$ 6 .

$a_3 =$ 

6×2
$+ 6 \times 2$
$+ 2^3$

$\therefore$ Must be $Y = 6$.

$= 4 \times 6 + 2^3 - 2 \times 6 - 2^3$
$= 2 \times 6$

(b) Jordan notices that a_2, a_5, and a_8 are all divisible by 5. Does this pattern continue? Carefully prove your answer.

a(2) = 3*i

a(5) = 3*j

a(8) = 3*k for some integers i, j, k

Prove that for all n >= 2, where n = n + 3*k, for some k as an integer, that

a(n) is divisible by 3.

Test n = 2 (already done)

Assume true for n = k
a(k) = 3*t, for some integer t.

Prove true for n = k + 3

a(k+3) = 2*a(k+2) + 6*a(k+1)

= 2*(2*a(k+1) + 6*a(k)) + 6*a(k+1)
= 10 * a(k+1) + 12 * a(k)
= 10 * (2*a(k) + 6 * a(k - 1)) + 12 * a(k)
= 32 * a(k) + 60 * a(k-1)
= 3 * (3t + 20*a(k-1))

Therefore true, divisible by 3.

(c) Jordan says, "I think the closed form solution for this recurrence relation should be of the form

$$a_n = A\left(\frac{2+\sqrt{28}}{2}\right)^n + B\left(\frac{2-\sqrt{28}}{2}\right)^n$$

for some integers A and B. But there should always be an integer number of towers for any height n, while this formula looks like it will usually produce irrational values. I must have made a mistake."

Has Jordan made a mistake? Briefly justify your answer.

No, not necessarily.

The root values within the polynomial expansion can simply be cancelled out for appropiate values of A and B.

For example, when n = 2:

a(n) = A(8 + sqrt(7)) + B(8 - sqrt(7)
If A = B, then the sqrt(7) is cancelled.

Question 5.

(a) Use mathematical induction to prove that for all $t \in \mathbb{N}$, the set $S = \{1, 2, 3, \ldots, t\}$ with t elements has exactly 2^t subsets.

Test for n = 0.
An empty set obviously only has one subset (itself)

Assume true for n = k.
|Subsets of S(k)| = 2^k

Prove true for n = k + 1
|Subsets of S(k+1)| = 2^(k+1)
= 2 * |Subsets of S(k)|

Therefore, true, as the inductive step hypothesis that there is a k -> 2^k mapping for elements to subsets; the definition of (k+1) -> 2 * 2^k is achieved.

(b) Consider a set S of size 17. Let $T \subseteq \mathcal{P}(S)$ be a set such that the elements of T are pairwise non-disjoint (that is, $A \cap B \neq \varnothing$ for all $A, B \in T$).
Construct an example showing that there exists such a set T whose cardinality is exactly 2^{16}.

Consider some arbitrary set, such that:
S = {1, 2, 3, 4, ..., 25}
The power set P(S) contains all subsets within S. If T = P(S), there would be disjoint elements, for example: {2}, {3, 4, 5}
Hence, fix one element as a mandatory element within the subsets.
Choose any arbitrary value, such as 1. Therefore, the subsets of this new set, would look something akin to:
{{1}, {1, 2}, {1, 3}, {1, 4}, ..., {1, 2, 3, ..., 25}}

In this set, the number of sets possible is 2^24, as we have fixed one - hence, we can choose to use (or not use) the 24 other elements. QED.

(c) Carefully prove that no such set T as defined in part (b) can have cardinality greater than 2^{16}.

Consider once again, the subsets that are created when there is an arbitrary element in T fixed within the subset.
To find other possibilities, divide P(S) into two distinct subcollections of P(S).
S(1), which contains all subsets that have 1 as an element.
S(~1), which contains all subsetes that don't have 1 as an element.
S(~1), may only contain elements within {2, ..., 100}. Hence, at most, |S(~1)| = 2^24.
Further consider that S(1) cannot contain any complement of S(~1), and hence, at most, contains 2^24 - |S(~1)|.

We may now conclude that at most, the set shown in part b) has at most 2^24 elements.

Question 6.

In the popular monster-collection game Pokumon, monsters can have different types: Acid, Baby, Cosmic, Dark, Evergreen, or Fighting, which for the rest of this question we will refer to as a, b, c, d, e, f. Certain types do bonus damage against certain other types, as summarised in the below "Type Chart". If there is a 1 in the row for Type x and the column for Type y, this means Type x does bonus damage against Type y. Otherwise, a 0 implies no bonus damage is done.

	a	b	c	d	e	f
a	0	1	0	1	1	1
b	1	0	0	1	1	0
c	1	1	1	1	1	1
d	1	0	0	0	0	0
e	1	1	0	0	1	0
f	0	0	1	1	1	1

Let R be the relation on the set of types $\{a, b, c, d, e, f\}$ defined by $x \ R \ y$ if and only if Type x does bonus damage against Type y.

(a) Find a set S of size at least 2 for which the relation R on S is reflexive.

c, e, f

(b) Find a set S of size at least 2 for which the relation R on S is symmetric.

a, b

(c) Find a set S of size at least 3 for which the relation R on S is transitive.

c, e, f

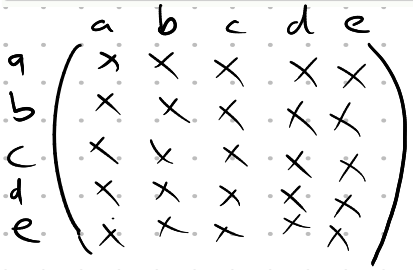
(d) The Type Chart above is converted to a 6×6 matrix M, and the matrix M^2 is calculated to be

$$M^2 = \begin{bmatrix} 3 & 1 & 1 & 2 & 3 & 1 \\ 2 & 2 & 0 & 1 & 2 & 1 \\ 4 & 3 & 2 & 4 & 5 & 3 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 2 & 2 & 0 & 2 & 3 & 1 \\ 3 & 2 & 2 & 2 & 3 & 2 \end{bmatrix}.$$

What can be deduced from the fact the $(3, 4)$th entry of M^2 is 4?

The number of length 2 "walks" from 3 to 4.

(e) For the sequel to Pokumon, the Type Chart is going to be redesigned. There will now be only 5 different monster types total, and the designers want each type to be able to do bonus damage to either 3 or 4 other types. How many possible Type Charts are there?



$$\left(\binom{5}{3} + \binom{5}{4}\right)^5$$

My wrong solution:

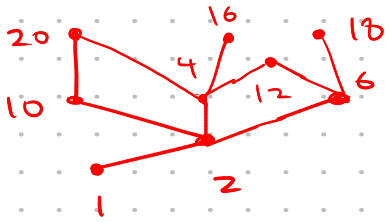
$$\left(\binom{5}{3}\right)^5 + \left(\binom{5}{4}\right)^5$$

This limit to every column having 3 or 4.
However, it should really give the choice of 3 or 4 each row.

Question 7.

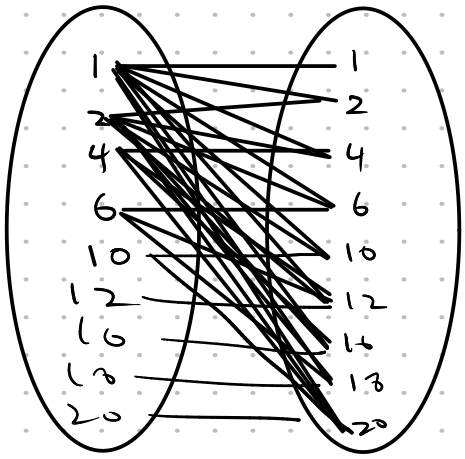
(a) Consider the set $S = \{1, 2, 4, 6, 10, 12, 16, 18, 20\}$.
How many edges are in the standard Hasse diagram for the partially-ordered set $(S, |)$?

10



How many edges are in the unsimplified diagram for $(S, |)$?

30



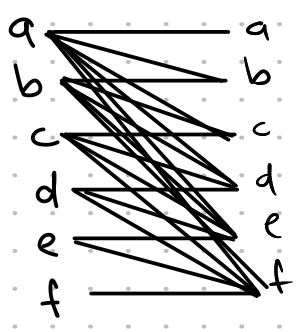
(b) Suppose T is a set with $|T| = 10$, and the standard Hasse diagram for a poset $(T, \preceq)$ has 8 edges, while its unsimplified diagram has 18 edges.
What can be deduced about the poset $(T, \preceq)$?

We could state a couple of things.
Due to the fact that the unsimplified diagram having far more edges than the Hasse diagram, the poset has multiple 'levels' or 'hiearchies' present within the Hasse diagram.

Further more, it is evident that not every single element within the poset is related to another, as the amount of edges within the Hasse diagram do not represent this.

(c) Suppose T is a set with $|T| = 6$. What is the maximum number of edges that can be in the unsimplified diagram for $(T, \preceq)$?

21



Note: Has to be anti symmetric, i.e. you can't have a≤b & b≤a.

(d) Suppose $|T| = 13$, and that the poset $(T, \preceq)$ has a greatest element. What is the minimum number of edges that can be in the unsimplified diagram for $(T, \preceq)$?

Minimum would be just reflexive.

1 ——— 1
2 ——— 2
3 ——— 3
4 ——— 4
⋮ ⋮

∴ Greatest element means that all other elements must point to a single one.
So you have:
13 + 12 = reflexive + greatest

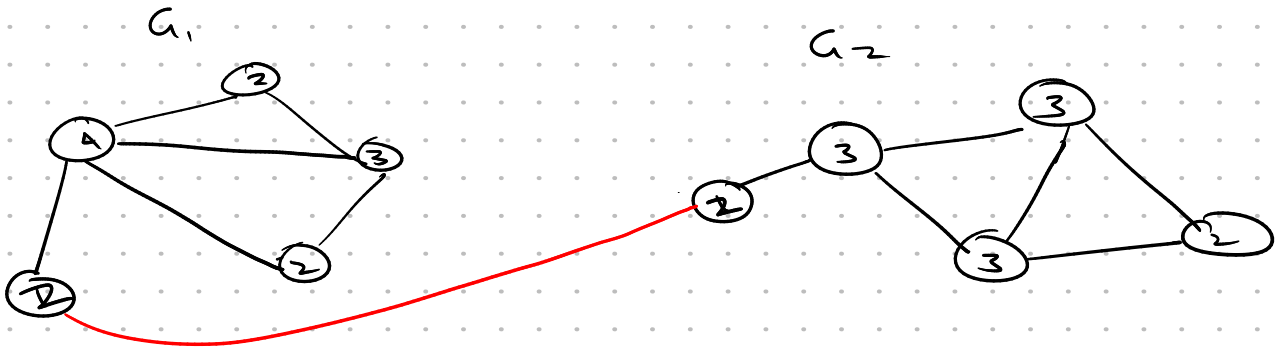
Question 6.

In Algorithm Archipelago, there are two islands that each have several towns connected by roads. These towns and roads form a connected graph on each island, labelled G_1 and G_2 respectively. A bridge is going to be built between two towns, one from each island, effectively introducing a new edge between one vertex in G_1 and one vertex in G_2. This new graph represented the joined islands will be called H.

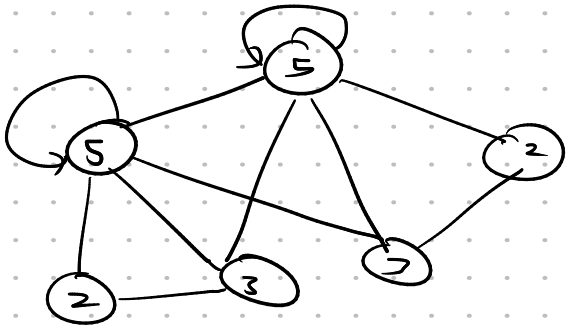
(a) Suppose G_1 has vertex degree sequence $4, 3, 2, 2, 1$ and G_2 has vertex degree sequence $3, 3, 3, 2, 1$. Find a vertex degree sequence for a new graph H which has no pendant vertices.

Enter your answer as a sequence of numbers separated by commas.

4, 2, 3, 2, 2, 2, 3, 3, 3, 2



(b) Explain why no matter what vertex degree sequences G_1 and G_2 have, H cannot have vertex degree sequence $5, 5, 3, 3, 2, 2$.



If the graph H is valid, then it should be possible, by removing one of the edges from the graph H, to create two connected components (or islands). Considering the vertex degree sequence, the two vertexes that could be possibly removed to create a partitioned connected components (the two 2's), are both connected to 5's and 3's. Hence, when removing an edge from one of these vertices, the entire graph stays connnected.

(c) For each statement below, select the correct conclusion.

"If both G_1 and G_2 contain a Hamilton cycle, then H contains a Hamilton cycle."

- ○ This statement is always true.
- ○ This statement is sometimes true and sometimes false. ✓
- ◉ This statement is always false.

Imagine two graphs with a Hamiltonian circuit are connected by a single edge. The single edge part of the problem, presents an impossibility for the hamiltonian circuit, as it must traverse through that edge twice to get back to the starting point.

"If both G_1 and G_2 are trees, then H is a tree."

- ◉ This statement is always true.
- ○ This statement is sometimes true and sometimes false. ✓
- ○ This statement is always false.

Joining up a tree at any node simply just creates a new subtree - hence, this is always true.

"If both G_1 and G_2 contain an Euler trail, then H contains an Euler trail."

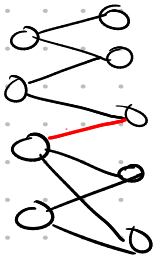
- ○ This statement is always true.
- ◉ This statement is sometimes true and sometimes false. ✓
- ○ This statement is always false.

Depends on the graphs vertex degree sequence, and as well, where the graphs were joined at. They must have an even degree for every single vertex.

"If both G_1 and G_2 are bipartite, then H is bipartite."

- ◉ This statement is always true.
- ○ This statement is sometimes true and sometimes false. ✓
- ○ This statement is always false.

Consider the two bipartite graphs below. Join any two bi-partite graphs, at any node, and then the arrival node simply becomes the opposing node - and the bipartite graphs have been combined.



(d) Mayor Tin has noticed that both G_1 and G_2 are planar. They are worried that since creating H adds a new edge but does not change the total number of vertices, that this will violate Euler's formula for planar graphs. Is their claim correct? Explain your answer.

Mayor Tin forgot that creating a new edge between G1 and G2 would remove a region from H, as the external regions of G1 and G2 would combine into one.
Hence, this becomes:
r - 1 - e + 1 + v = 2
r - e + v = 2

Question 8.

Jasper is playing a role-playing game in which they can allocate skill points to any of 6 different abilities. A player always starts with 0 skill points in each ability, and each time they earn 1 skill point, they assign it to their choice of ability. Each ability can have a **maximum** of 100 skill points assigned to it.

(a) Jasper announces that they have assigned a total of 59 skill points. In how many ways could they have done this?

comb(64, 5)

$a_1 + a_2 + a_3 + \dots + a_6 = 59 \quad = \binom{64}{5}$

(b) Jasper's more experienced friend Oliver announces that they have assigned a total of 555 skill points. In how many ways could they have done this?

comb(50, 5)

$a_1 + a_2 + a_3 + \dots + a_6 = 555.$ $a_i \leq 100$ not satisfied

$\therefore 100 - b_1 + 100 - b_2 + \dots + 100 - b_6 = 555$

$\therefore b_1 + b_2 + \dots + b_6 = 45. \quad = \binom{50}{5}$

(c) Jasper's friend Charles has assigned a total of 192 skill points. Charles reports the following facts:

- They have assigned a skill points to the Archery ability.
- The Archery ability has the least amount of skill points assigned to it.
- Each other ability has at most $a + 8$ skill points assigned to it.

What is the minimum possible value for a?

26

What is the maximum possible value for a?

32

$a + b + c + d + e + f = 192.$

$a \leq b, c, d, e, f \leq a + 8.$

If evenly spread, then $a, b, c, d, e, f = 32$.

$\therefore \max(a) = 32.$

$\therefore \min(a) = 26$

Strategy
Maximum: Find where all the elements are equal, or atleast close to as equal as possible.

Minimum: Find the smallest value of a, such that every other value is equal. I did this by starting at x - 8 + x + ... = 192, and then slowly incremented up to -6, where I found everything was equal.

(d) To complete the game, Jasper must pass two "skill checks". In particular, they need:

- At least 12 skill points in the Archery ability or at least 18 skill points in the Buoyancy ability, **and**
- At least 16 skill points in the Camouflage ability or at least 10 skill points in the Dexterity ability.

Suppose at this stage that Jasper has assigned a total of 63 skill points, and the points have been assigned in such a way that Jasper is able to complete the game. In how many ways could they have assigned the skill points?

|(A U B) and (C U D)| = |A U B| + |C U D| = |A U B U C U D|

|A U B| = |A| + |B| = |A and B|

|A|:
A + B + C + D + E + F = 63
A + B + C + D + E + F = 51 = comb(55, 5)

|B|:
A + B + C + ... = 63 - 18 = 45
A + B + C + ... = comb(50, 5)

|C|:
A + B + C + ... = 63 - 16 = 47
A + B + C + ... = comb(52, 5)

|A| = comb(55, 5)

|D|:
A + B + C + ... = 63 - 10 = 53

|A and B|:
A + B + C + D + E + F = 63 - 30 = 33 => comb(38, 5)

|A and C|:
A + B + C + ... = 63 - 28 = 35 => comb(40, 5)

|A and D|:
A + B + C + ... = 63 - 22 = 41 => comb(46, 5)

|B and C|:
A + B + C + ... = 63 - 34 = 29 => comb(34, 5)

|B and D|:
A + B + C + ... = 63 - 28 = 35 => comb(40, 5)

|C and D|:
A + B + C + ... =

ahhh case bash cringe and unbased noon cares

Question 10.

(a) Ida begins with just one weapon that does 38 damage each turn. Their first foe (i.e. enemy) is the snoring snapdragon. It has 281 health, and never attacks back, but it only perishes (i.e. dies) when its health reaches exactly 0. If it suffers an attack that would leave it with negative health, instead of perishing, it regenerates 95 health. For example, if the snapdragon has 8 health and receives 10 damage, its health will be reduced to -2 and its regeneration spell will regain 95 health, leaving it with 93 health before the next turn.

Is Ida able to defeat the snoring snapdragon? Why or why not?

No;

Consider the following sequence:
281 - 38 * 8 => -23 + 95

72 - 38 * 2 => -4 + 95
91 - 3 => -23 + 95

A loop occurs

The reason for this is, is that the mod of gcd(95 % 38, 38) != 1, and therefore, loops can occur after a certain amounts of steps.

Select all weapons that would allow an adventurer to eventually defeat the snoring snapdragon.

- [x] The denim dagger, which does 66 damage each turn.
- [ ] The silver sword, which does 75 damage each turn.
- [x] The blue blade, which does 33 damage each turn.
- [ ] The mauve mace, which does 25 damage each turn.
- [ ] The crimson cutlass, which does 57 damage each turn.

(b) Ida's second foe is the hale hippogriff. It has 2808 health, and never attacks back, but it also only perishes when its health reaches exactly 0. If it suffers an attack that would leave it with negative health, instead of perishing, its health resets to 2808. This time Ida has two weapons at their disposal: the vorpal sword which does 84 damage in one turn, and the manxsome blade which does 66 damage in one turn. Ida claims that they are able to defeat the hale hippogriff. What is the least number of attack turns required to defeat the hale hippogriff?

Solve for the equation 84*x + 66*y = 2808

Make sure to divide through if there is a gcd:
14*x + 11*y = 468

Solving this equation:
14 = 11 + 3
11 = 3 * 3 + 2
3 = 2 + 1
Reversing:
1 = 3 - 2
1 = 3 - 11 + 3 * 3
= 4 * 3 - 11
= 4 * (14 - 11) - 11
= 4 * 14 - 5 * 11
Therefore:
368 = 1472 * 14 - 1840 * 11
We have:
x = 1472 + 11*t
y = -1840 + 14*s

Minimise one of them (let's choose x)
x = 1472 + 11 * (-133)
y = -1840 + 14 * (133)
x = 9 y = 22
Therefore, minimum is 31

Now try minimising y:
y = -1840 + 14 * 132
x = 1472 + 11 * (-132)
x = 20 y = 8
Therefore, minimum is 28

Hence, the minimum is 28.

(c) Leonardo's third foe is the retaliatory wraith, but it is undefeatable! The retaliatory wraith has three weapons at its disposal, and on each turn will use exactly one of them. The weapons are:

- the small sabre, which does 19 damage in one turn,
- the middling mace, which does 20 damage in one turn, and
- the large lance, which does 21 damage in one turn.

The retaliatory wraith decides to draw up a battle plan, listing which choice of weapon it will use each turn until Leonardo, who has 62 health, perishes (once Leonardo's health reaches a non-positive amount). How many such battle plans exist?

73

First, acknowledge that any attack combinations > 4 are impossible, as the monster is already dead.
Consider 3 turns:
The only possible combinations are:
LLM and LLL.
This is comb(3, 2) + 1 = 4

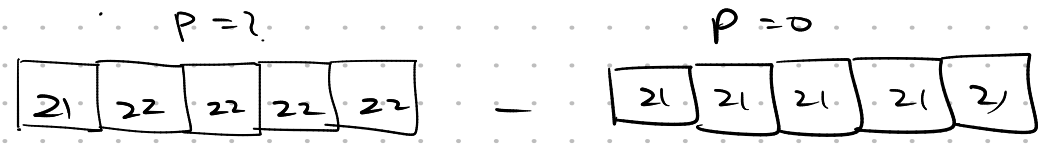
Consider 4 turns:
We can hypothesise that any single combination is valid, besides the ones that start with LLM and LLL, as that would be futile.
Hence, we have
3^4, for the amount of combinations overall, and then
- (comb(3, 2) * 3 + 3), for the different ways we can construct 4 turn attack moves that start with LLM and LLL.

Hence overall, we have:
4 + 3^4 - (comb(3, 2) * 3 + 3)

Question 11

(a) Your first guess is PQAJI, and the response is that P is in the secret word, but not in the first place, and Q, A, J, I are not in the secret word. How many possibilities remain for the secret word?

21 * 22^4 - 21^5



Everywhere else, we don't know.

$= P \geq 1.$

(b) While you are still thinking about the puzzle in (a), a friend who has already solved the same puzzle tells you that the secret word has no repeated letters. Now how many possibilities remain for the secret word?

* perm(21, 4) - perm(21, 5)

Same as above, but with permutations.

← no duplicates

$21 \times {}^{21}P_4 - {}^{21}P_5.$

(c) Next day you are playing the game again, so that the secret word may be the same or may be different. This time your first guess VZSKL is very lucky and you are told that V is in the secret word but not in the first place; Z is in the secret word but not in second place; S is in the secret word but not in third place; K is in the secret word but not in fourth place; and L is not in the secret word. Your friend again tells you that there are no repeated letters. How many possibilities are there for this day's secret word?

Let U be the set of lucky words which contain VZSK but don't contain L.
|U| = 21 * 5! (re-arranging the letters)

Let S_(V, Z, S, K) be the set of lucky words that contain VZSK, doesn't contain J, and has V in first place.
21 * 4!

Let S_V and S_Z, S_V and S_S .... be the set of lucky words that contain VZSK that don't contain J, and have the two letters at the respective
21 * 3!

Let S_V and S_Z and S_S, ... be the set of lucky words that contain VZSK that don't contain j, and have the three letters at their respective conditions
21 * 2!

Let S_V and S_Z and S_S and S_K:
21

Therefore, combinations are

21 * 5! - (4 * 21 * 4! - C(4, 2) * 21 * 3! + C(4, 3) * 21 * 2! - 21)

(d) Finally one more day. Again you are told that there are no repeated letters. From your first two guesses you determine that the secret word contains I, U and E; that it does not contain any of B, M, A, T, P, F; that I is not in the first or the second place, U is not in the third place and E is not in the fourth place. How many possibilities remain for this secret word?

S_I(1), S_I(2), ... = C(17, 2) * 4!
S_I(1) and S(U) = C(17, 2) * 3!
S_I(1) and S(U) and S(E) = 2 * C(17, 2) * 2!

C(17, 2) * 5! = Chose two other alphabetic characters, and arrange characters into positions.

C(4, 1) * 4! * C(17, 2) = 4 different found characters to fix to their positions, 4! different positions for other characters, 2 different other alphabetic characters.

(C(4, 2) - 1) * 3! * C(17, 2) = C(4, 2) - 1 represents every combination of two found positions, besides the double up of I (I cannot exist twice), 3! different positions and two other alphabetic characters

2 * 2 * C(17, 2) = 2 combinations of three in fixed place (C(4, 3) - 2, as we cannot have I double up), 2 different positions for remaining characters, chose two other alphabetic characters.
C(17, 2) * 5! - (C(4, 1) * 4! * C(17, 2) - (C(4, 2) - 1) * 3! * C(17, 2) + 2 * 2 * C(17, 2)) = 6800

Question 12.

For any positive integer n, we define the set

$$S_n = \left\{ k \in \mathbb{Z} \mid 3n + 4 \le k \le 3n^2 + 8 \right\}.$$

Find the number of elements in each of the following sets.

Syntax advice: You may give your answers as exact numbers, or as expressions in Maple notation. As well as the standard operations +, -, *, /, ^ you may also write

- `n!` for factorials,
- `c(n,r)` for combination numbers (binomial coefficients), and
- `p(n,r)` for permutation numbers.

Make sure you include any necessary brackets. Always use * to denote multiplication.

(a) If $A = \{ S_n \mid 6 \le n \le 26 \}$, then the number of elements in A is

21 .

$S_6, S_7, S_8, \ldots, S_{26}$ = 21 sets.

(b) If $B = \{ x \mid x \in X \text{ for some } X \in A \}$, then the number of elements in B is

3*(26)^2 + 8 - 3 * (6) - 4 + 1 .

Since $3n+4 \le k \le 3n^2+8$

$\Rightarrow (3 \times (26^2) + 8) - (3 \times (6^2) + 4) + 1$

(c) If $C = \{ x \mid x \in X \text{ for some } X \subseteq A \}$, then the number of elements in C is

21 .

Same idea as above.

(d) If $D = \{ x \mid x \in X \text{ for all } X \in A \}$, then the number of elements in D is

3 * (6)^2 + 8 - 3 * 26 - 4 + 1 .

(e) If $E = \{ x \mid x \in X \text{ for all } X \subseteq A \}$, then the number of elements in E is

0 .

such an X does not exist.

(f) If $F = \{ \text{functions from } S_6 \text{ to } S_7 \}$, then the number of elements in F is

.

$|S_7| = (3 \times 7^2 + 8) - (3 \times 7 + 4) + 1$

$|S_6| = (3 \times 6^2 + 8) - (3 \times 6 + 4) + 1$

To find elements from n → b, it is b − n + 1.

To make functions from S_6 to S_7, we must find every combination of one S_6 value mapping to one or more S_7 values.

$= 131^{95}$

(g) If

$$G = \{ \text{one-to-one (injective) functions from } S_6 \text{ to } S_7 \},$$

then the number of elements in G is

perm(131, 95) .

(h) If

$$H = \{ \text{onto (surjective) functions from } S_6 \text{ to } S_7 \},$$

then the number of elements in H is

0 .

Question 13.

(i) If R is reflexive or S is reflexive, then T is reflexive.

- (•) True
- ( ) False ✓

Since R or S are reflexive, then R must contain 9 R 9, 11 R 11, ... or S must contain 9 S 9, 11 S 11, ...
Hence, T must also be reflexive, as the same mappings exist within T.

(ii) If T is reflexive, then either R is reflexive or S is reflexive.

- ( ) True
- (•) False ✓

If T is reflexive, it must contain 9 T 9, 11 T 11, 14 T 14, 20 T 20.
However, these relations are not necessarily gained from exclusively R or S.
Hence, T could gain these relations like:
9 R 9, 11 R 11 and 14 S 14, 20 S 20

(iii) If T is antisymmetric, then both R and S are antisymmetric.

- (•) True
- ( ) False ✓

If T are anti-symmetric, then each relation that is contained within T has been obtained from R and S.
Consider the relations of T in a set S.
Hence, set S is partitioned into two sets - obtained from R and S.
Hence, since these two subsets are also anti-symmetric, R and S must also be anti-symmetric.

(b) Let P be a relation on a set B. We define a relation Q on B by

$$x \; Q \; y \quad \text{if and only if} \quad \text{there exists } z \in B \text{ such that } x \; P \; z \text{ and } z \; P \; y$$

(i) If

$$B = \{3, 14, 15, 20\} \quad \text{and} \quad P = \{(20, 3), (3, 3), (3, 14), (14, 15)\},$$

find the relation Q, giving your answer as a set of ordered pairs.

Syntax advice: You must use Maple notation; specifically, square brackets for ordered pairs whose entries are separated by commas, and curly brackets for the set whose elements are separated by commas. For example, if the answer is

$$Q = \{(1, 2), (3, 4), (5, 6)\},$$

then enter

```
{[1,2],[3,4],[5,6]}
```

$Q =$ I, 3], [20, 14], [3, 14], [3, 3], ✓

{[20, 3], [20, 14], [3, 14], [3, 3], [3, 15]} (dont forget reflexive relation!)

(ii) For relations P and Q, we say that $Q \subseteq P$ when the set of ordered pairs comprising Q is a subset of the set of ordered pairs comprising P.

Prove that P is transitive **if and only if** $Q \subseteq P$.

First prove that if Q is a subset of P then P is transitive.
If Q is a subset of P, then every ordered pair within Q exists within P. By definition of Q, it must be true that in every equation, xPz and zPy, for an ordered pair within Q. However, since Q is a subset of P, it must be true that xPy and yPy for some ordered pair (x, y) in Q, which also satisfies the transitivity of P.

Prove that if P is transtivie, then Q exists within P.
If P is transitive, then for every ordered element xRy and yRz, (x, z) exists. Since this is also the innate definition of Q, it follows that Q is indeed a subset of P.

(iii) By considering the example where $B = \mathbb{Z}$ is the set of integers and $x \; P \; y$ means $x < y$, or otherwise, prove that the following statement is **not true**: if P is transitive then $Q = P$.

Assume that if P is transitive, then Q does indeed = P.
Hence, when B = Z, and P is some set, for example = {(1, 2), (2, 3), (1, 3)}, we get Q = {(1, 3)}. However, immediately a contradiction is shown. P is transitive, but Q != P. Hence, the statement is false.

Question 14.

(a) The aim of this part is to find a solution of the congruence

$$x^2 \equiv -568 \pmod{1231^2}.$$

Let $x = 66 + 1231t$, where t is an integer. Then

$x^2 \equiv -568 \pmod{1231^2}$ if and only if $at \equiv b \pmod{1231}$

for certain integers a and b.

Find suitable values of a and b and enter them in the box **in that order**.

Syntax advice: Enter your answer as an **ordered pair** using square brackets and separated by a comma. For example, enter the values $a = 12$ and $b = -34$ using the syntax

```
[12,-34]
```

Answer: suitable values of a and b are [132, -4] .

Before attempting to solve the congruence $at \equiv b \pmod{1231}$, it is wise to confirm that it actually does have a solution. This is true because (choose the correct reason)

- (•) a and 1231 are relatively prime
- ( ) b and 1231 are relatively prime
- ( ) 1231 is a factor of b
- ( ) a, b and 1231 are relatively prime
- ( ) a and b are relatively prime
- ( ) 1231 is not a factor of b
- ( ) b is a multiple of a

$(66 + 1231t)^2 \equiv -568 \pmod{1231^2}$

$(1231)^2 t^2 + 2(1231)(66)t + 66^2 \equiv -568 \pmod{1231^2}$

$= 0 + 2(1231)(66)t \equiv -4924 \pmod{1231^2}$

$= 132t \equiv -4 \pmod{1231}$

By solving the congruence $at \equiv b \pmod{1231}$, find **one** suitable value of t.

(You do not need to give all values of t: enter your answer as a single number.)

Answer: $t \equiv$ 1604 $\pmod{1231}$.

$132t \equiv -4 \pmod{1231}$

$1231s + 132t = 1$

$1231 = 132 \times 9 + 43$

$132 = 43 \times 3 + 3$

$43 = 3 \times 14 + 1$

$1 = 43 - 3 \times 14$

$1 = 43 - 14 \times (132 - 43 \times 3)$

$= 43 - 14 \times 132 + 42 \times 43$

$= 43 \times (1231 - 132 \times 9) - 14 \times 132$

$= 43 \times 1231 - 387 \times 132 - 14 \times 132$

$1 = 43 \times 1231 - 401 \times 132$

$-4 = -172 \times 1231 + 1604 \times 132$

Hence, find **one** solution of the congruence $x^2 \equiv -568 \pmod{1231^2}$.

(You do not need to give all values of x: enter your answer as a single number.)

Answer: $x \equiv$ (66 + 1231 * 1604) $\pmod{1231^2}$.

(b) By modifying the ideas used in (a), or otherwise, prove that for every positive integer n, the congruence

$$x^2 \equiv -568 \pmod{1231^n}$$

has a solution.

$(66 + 1231t)^2 \equiv -568 \pmod{1231^n}$

$\Rightarrow 66^2 + 2 \times 66 \times 1231t + (1231)^2 t^2 \equiv -568 \pmod{1231}$

$\Rightarrow 132t \equiv -4 \pmod{1231^{n-2}}$

$\therefore$ Always has solution, 1231 is prime, and 132 is not a multiple of 1231, $\therefore$ always coprime, $\therefore$ always solution.

Question 15.

A metafield is an object studied in the field of mathematical stereotopodynamics. A metafield may be doubly-Euclidean, or not, and it may be sporadic, or not.

Let X be a metafield. We wish to prove the statement "if X is doubly-Euclidean, then X is sporadic".

Select all of the following proof outlines which could **not possibly** give a proof of the above statement. That is, if there is no possibility of filling in the dots in an option so as to give a logically correct proof of the statement, then you should select it; if it is possible to fill in the dots so as to obtain a correct proof, then you should not select it.

- [x] Suppose that X is both doubly-Euclidean and sporadic ..... This is a contradiction.
- [ ] Suppose that X is not sporadic ..... Therefore X is not doubly-Euclidean.
- [ ] Suppose that X is doubly-Euclidean ..... Therefore X is not doubly-Euclidean.
- [x] Suppose that X is not doubly-Euclidean ..... Therefore X is not sporadic.
- [ ] Suppose that X is doubly-Euclidean ..... Therefore X is sporadic.
- [ ] Suppose that either X is doubly-Euclidean or X is sporadic ..... This is a contradiction.
- [ ] Suppose that X is doubly-Euclidean and not sporadic ..... This is a contradiction.
- [x] Suppose that X is doubly-Euclidean ..... Therefore X is not sporadic.

Question 16.

The graph G has 10 vertices, and the degrees of its vertices are

$$6, 5, 5, 5, 5, 3, 2, 2, 2, 1 .$$

Prove that G is **not** bipartite.

Be sure to give a clearly written, detailed and logically accurate answer - full marks will not be given for sketchy work.

For the graph G to not be bipartite, it must be able to be split into two distinct sets, such that edges only go between those sets. Hence, consider adding any arbitrary vertex, and then creating some properties about two sets.

Therefore, define two sets, S and T, and add 8 to S.

Therefore, there must be atleast 7 vertices in set T, that are adjacent to the vertex 8.
Therefore, there must only be 3 vertices in A.

Therefore, all vertices with a degree higher than 3, must be in A.

However, this means v1, v2, v3, v4 and v5 must be in A, which is a contradiction..

Therefore, G is not bipartite.